

Set-1

1. Choose True/False for the following statements.

$1 \times 5 = 5$

- (a) A *strictly dominant* strategy may not be unique.
- (b) When a strategic form game has *multiple* pure strategy Nash equilibria (PSNE), the payoffs for a player can be different in different PSNEs.
- (c) In a pure strategy Nash equilibrium (PSNE), the equilibrium strategy of each player offers a best response (BR) irrespective of the strategies of the rest of the players.
- (d) A game with a finite number of players and strategies has *at least one* Mixed Strategy Nash Equilibrium (MSNE).
- (e) In MSNE, the expected utility of a player i must be same for any strategy that has probability $(0, 1]$.

2. Find PSNE for the following game:

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		P-2			
		A	B	C	D
P-1	A	5, 2	2, 6	1, 4	0, 4
	B	0, 0	3, 2	2, 1	1, 1
	C	7, 0	2, 2	1, 5	5, 1
	C	9, 5	1, 3	0, 2	4, 8

3. Assume two bidders with valuations v_1 and v_2 for an object. Their bids are in multiple of some unit (that is, discrete). The bidder with higher bid wins the auction and pays the amount that he has bid. If both bid the same amount, one of them gets the object with equal probability 0.5. Find a PSNE of this game, if exists.

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4. (a) Write down the necessary and sufficient conditions such that a mixed strategy profile $(\sigma_1^*, \dots, \sigma_n^*)$ is a MSNE.
- (b) Using the necessary and sufficient conditions for an MSNE, investigate if there exists an MSNE with the *support* $\{A, B\} \times \{A, B\}$.

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		P-2	
		X	Y
P-1	A	20, 0	0, 10
	B	0, 90	20, 0

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5. Apply iterative elimination of strongly dominated strategies to the following problem and show the reduced matrix in each iteration. 5

		P-2		
		X	Y	Z
P-1	A	5, 1	2, 2	1, 3
	B	2, 3	5, 4	6, 0
	C	3, 2	3, 0	0, 4

6. Suppose a team of n UAVs (unmanned aerial vehicles) $r_1, r_2, \dots, r_n$ are engaged in surveillance of a workspace W , in a coordinated way. The UAVs are configured (i.e., positioned) in W in a way that they maximize the *coverage area* collectively. Assume that, at some time point t , the UAVs are configured in positions $\{p_1, p_2, \dots, p_n\}$ in a way that they maximally cover W . At time $t + 1$, one of the UAVs r_i is made non-functional somehow, making a partial void in the surveillance task. Now, answer the following:

- An UAV r_k from the remaining $n - 1$ functional UAVs move close to p_i (the position of r_i) to compensate the void. Suggest a technique how the functional UAVs unanimously choose an r_k for this task in a *decentralized way*. 5

7. Suppose in a disaster-hit area, a target located at p_d is waiting to be served by a ground robot located at p_r . There are many paths $\{\pi_i\}$ between p_r and p_d . You have to do a tri-criteria search with the following *weighted* criteria:

- (i) Path cost $c(\pi_i)$ of the robot with weight $\alpha_1 \geq 0$,
- (ii) Wait time $\omega(\pi_i)$ of the target with weight $\alpha_2 \geq 0$,
- (iii) A fixed hard deadline D , by which the target must be served, with weight $\alpha_3 \geq 0$

Note that $\alpha_1 + \alpha_2 + \alpha_3 = 1$. Now, answer the following:

- Formulate strict dominance and weak dominance. 5